

1. Administrative & Study Identification

Full Study Title: A Phase 1/1b Open-label, Multicenter Clinical Study of MK-0472 as Monotherapy and Combination Therapy in Participants With Advanced/Metastatic Solid Tumors **Short Title/ Acronym:** Study of MK-0472 in Participants With Advanced/Metastatic Solid Tumors (MK-0472-001) **Protocol Number:** MK-0472-001 **NCT Number:** NCT05853367 **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** MK-0472 (SHP2 inhibitor) [Monotherapy and in combination with Pembrolizumab (MK-3475) or MK-1084] **Phase of Development:** Phase 1 (Phase 1/1b) **Medical Monitor:** Clinical Director, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy, safety, and tolerability of MK-0472 administered as monotherapy and in combination with pembrolizumab or MK-1084 in participants with advanced/metastatic solid tumors, evaluating specifically Dose Limiting Toxicities (DLTs), Adverse Events (AEs), and treatment discontinuations due to AEs. **Secondary Objectives:** To evaluate the pharmacokinetics (PK) of MK-0472, including maximum concentration (C_{max}), trough concentration (C_{trough}), and area under the concentration-time curve (AUC_{tau}).

2.2 Background & Rationale Patients with advanced or metastatic solid tumors who have progressed on or are intolerant to standard therapies have a high unmet medical need. This study investigates the safety profile and initial efficacy of MK-0472, a novel SHP2 inhibitor, both alone and in combination with a PD-1 inhibitor (pembrolizumab) or a KRAS G12C inhibitor (MK-1084) to overcome resistance mechanisms and enhance anti-tumor activity in targeted populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Recruiting (Currently recruiting participants) **Control Type:** Single-arm/Uncontrolled (Multiple experimental cohorts) **Blinding:** Open-label **Randomization:** N/A (Dose escalation/cohort expansion)

3.2 Planned Enrollment & Duration Target N\$: 178 participants **Number of Sites:** Multicenter (Global sites including US, Canada, Spain, Switzerland, Israel, and Poland) **Estimated Study Start:** 06-Jul-2023 **Estimated Primary Completion:** 12-Feb-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histologically or cytologically confirmed diagnosis of advanced/metastatic solid tumors. 2. Has received, or been intolerant to, all available treatment known to confer clinical benefit. 3. Specific infectious disease statuses permitted: HIV well-controlled on stable ART; HCV undetectable; HBsAg positive with at least 4 weeks HBV antiviral therapy and undetectable viral load. 4. Meets cohort-specific criteria: - **Arm 1:** Oncogenic receptor tyrosine kinase (RTK) pathway alterations. - **Arm 2:** Tumor types known to be sensitive to anti-PD-1/L1 therapies. - **Arm 3:** Histologically or blood-based confirmation of KRAS G12C mutation.

4.2 Exclusion Criteria 1. Known hypersensitivity to MK-0472, MK-1084, or pembrolizumab, or any of their excipients. 2. Received therapy with a proton-pump inhibitor (PPI) or an H2 histamine blocker receptor antagonist within 7 days before the first scheduled day of dosing. 3. Has discontinued prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent due to an adverse event. 4. Clinically significant cardiovascular disease or specified ophthalmological findings (e.g., intraocular pressure >21 mm Hg, central serous retinopathy). 5. Has active autoimmune disease requiring systemic treatment in the past 2 years (except replacement therapy) or history of/current pneumonitis/interstitial lung disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - MK-0472: Oral capsule daily per protocol-specified dose level. - MK-1084 (in combination arm): Oral capsule daily. - Pembrolizumab (Trade name: Keytruda, ATC code: L01FF02, [EMA Product Information](#)) (in combination arm): 200 mg administered on Day 1 of each 3-week (21-day) cycle. **Route of Administration:** Oral (MK-0472, MK-1084) and IV Infusion (Pembrolizumab). **Duration of Treatment:** Until disease progression, withdrawal/discontinuation, or up to 35 cycles (approximately 2 years) for pembrolizumab.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care for symptomatic treatment and well-controlled HIV, HBV, or HCV therapies. **Prohibited:** Systemic anticancer therapy within 4 weeks prior; proton-pump inhibitors (PPIs); H2 histamine blockers; live/live-attenuated vaccines within 30 days before the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Genetic/Tumor testing (RTK, KRAS, PD-L1), Viral

testing | | **Cycle 1** | Day 1 to 21 | First Dose, PK Sampling, Safety Assessment, DLT Evaluation | | **Ongoing Visits** | Every 21 Days | Treatment administration (Q3W), Safety Labs, Efficacy Assessment, AE Monitoring | | **End of Study** | Follow-up | Final Safety Assessment, Survival and Disease Progression Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number of Participants Who Experience a Dose Limiting Toxicity (DLT) as Assessed Using Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0 at the end of Cycle 1 (21 days). 2. Number of Participants Who Experience One or More Adverse Events (AEs) over up to approximately 56 months. 3. Number of Participants Who Discontinue Study Treatment Due to an Adverse Event (AE).

Measurement Method: Clinical observation, vital signs, and safety laboratory tests evaluated via CTCAE v5.0.

7.2 Secondary & Exploratory Endpoints - Pharmacokinetics

(\$C_{trough}\$): Lowest Plasma Concentration of MK-0472 reached before the next dose. - **Pharmacokinetics (\$AUC_{\tau}\$):** Area Under the Concentration Time-curve From Time 0 to the End of the Dosing Period of MK-0472. - **Pharmacokinetics (\$C_{\max}\$):** Maximum Plasma Concentration of MK-0472.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs evaluated using CTCAE v5.0 criteria throughout the duration of the study, up to 56 months.

Serious Adverse Events (SAEs): Evaluated concurrently with standard safety and toxicity profiles and reported to the sponsor. **Data Monitoring Committee (DMC):** Sponsor-directed internal review committee overseeing dose-escalation decisions and DLTs.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (All participants who received at least one dose of study treatment) for DLT/AE evaluation. PK population for concentration assessments. **Sample Size**

Justification: Target enrollment of 178 participants. Sample size is not driven by strict statistical power but is based on standard Phase 1 dose-escalation and cohort expansion rules (e.g., assessing DLTs to identify maximum tolerated dose / recommended phase 2 dose). **Handling of Missing Data:** PK and safety evaluations primarily utilize observed data; standard early-phase clinical trial imputation protocols apply where necessary.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Ongoing clinical monitoring to verify the safety and integrity of data across all global participating sites.

Audit Trail: Case Report Form (eCRF) electronic data capture with full audit trail features as per Merck clinical standards.

11. Study Identifiers & Governance

Primary ID: MK-0472-001 **Posting ID:** 1119893654167850156 **Registry Number:** NCT05853367 **Sponsor/Lead Organization:** Merck **Study Title:** Study of MK-0472 in Participants With Advanced/Metastatic Solid Tumors (MK-0472-001) **Official Regulatory Title:** A Phase 1/1b Open-label, Multicenter Clinical Study of MK-0472 as Monotherapy and Combination Therapy in Participants With Advanced/Metastatic Solid Tumors

12. Clinical Framework (Oncology/Solid Tumor Specific)

Primary Conditions: Metastatic Solid Tumors, Advanced Solid Tumors **Preferred UMLS Name:** Metastatic solid tumor **Disease Codes:** MeSH Code: D009369 | SNOMED ID: 263650009 **Diagnosis Threshold:** Histologically or cytologically confirmed disease with specific genetic markers (RTK pathway, KRAS G12C, or PD-1/L1 sensitive) **Medical Methodology:** SHP2 targeted therapy combined with immunotherapy (PD-1 checkpoint inhibition) or KRAS targeted therapy **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose (RP2D) with manageable toxicity

13. Study Procedures & Methodology

Management Pathway: Open-label, dose-escalation and combination cohort expansions **Education Protocol (HCP):** Investigator training on administration protocols, specifically managing DLT reporting in Cycle 1 **Education Protocol (Patient):** Informed consent detailing dosing compliance (daily oral capsules vs 21-day IV infusions) and immediate AE reporting **Quality Audit Mechanism:** Routine source data verification, centralized data review, and dose-escalation clearance meetings

14. Observation & Follow-Up Schedule

Total Duration: Up to ~56 months (Treatment up to ~2 years, plus post-treatment follow-up) **Onsite Visit Cadence:** Every 21 days (corresponding to 3-week cycles) **Remote Monitoring Frequency:**

Follow-up according to Phase 1 oncology clinical standards
Checklist Review Interval: Continuous AE monitoring evaluated extensively at the end of Cycle 1 and prior to subsequent cycles

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				